

# Design and Simulation of a Quaternion–PD Reaction-Wheel Attitude Control System with Disturbance Rejection

## ABSTRACT

Satellite attitude control is a critical subsystem in space missions, as the success of communication, Earth observation, navigation, and scientific payload operations depends directly on precise spacecraft orientation. With the growing use of small satellites and CubeSats, there is an increasing demand for attitude control systems that are reliable, computationally efficient, and capable of handling large-angle maneuvers under realistic disturbance conditions. Reaction wheels continue to be one of the most widely used actuators for such missions due to their high pointing accuracy and continuous torque capability. However, reaction-wheel-based attitude control systems face challenges related to nonlinear rotational dynamics, external disturbances, actuator saturation, and momentum accumulation, particularly during aggressive maneuvers.

This work proposes the development and simulation-based evaluation of a quaternion-based proportional–derivative (PD) attitude control framework for a rigid satellite equipped with a four-reaction-wheel pyramidal configuration. The proposed study aims to establish a complete end-to-end control architecture that integrates spacecraft rotational dynamics, quaternion kinematics, actuator torque allocation, and disturbance modelling within a unified simulation environment.

Quaternions are selected as the attitude representation to avoid singularities associated with Euler angles and to enable smooth and globally valid attitude tracking during large-angle reorientation maneuvers.

The satellite is planned to be modelled as a rigid body in low Earth orbit, with rotational dynamics derived using angular momentum conservation principles. The interaction between the spacecraft body and the internal reaction wheel assembly will be explicitly included in the model to accurately capture momentum exchange and coupling effects. Quaternion kinematic equations will describe the evolution of spacecraft orientation, while body angular velocity dynamics will be used to evaluate stability, transient response, and steady-state behavior. External disturbance torques, such as gravity-gradient effects and low-amplitude sinusoidal disturbances defined in the inertial reference frame, are intended to be incorporated and transformed into the body frame using quaternion-based rotation matrices.

A quaternion-error-based PD controller is proposed to generate the commanded body-axis torques required for attitude correction. The controller structure will utilize both attitude error and angular rate error to ensure convergence to the desired orientation while providing sufficient damping.

Controller gains are planned to be selected using classical second-order system design principles, allowing intuitive tuning through desired natural frequencies and damping ratios. This approach is expected to balance simplicity, robustness, and computational efficiency, making it suitable for implementation on resource-constrained satellite platforms.

To improve control authority and fault tolerance, a four-reaction-wheel pyramidal configuration is proposed instead of the conventional three-wheel arrangement. The redundant actuator layout is expected to provide improved robustness against wheel saturation and potential actuator degradation. A geometry-based torque allocation scheme using a pseudo-inverse mapping is planned to distribute commanded body torques among the individual reaction wheels while minimizing overall control effort. The influence of the pyramidal tilt angle on reaction wheel momentum accumulation and actuator usage is identified as a key design parameter and will be

systematically investigated in the proposed study.

The planned simulation campaign will include large-angle rest-to-rest slew maneuvers representative of typical mission scenarios. Performance metrics such as attitude tracking accuracy, angular rate convergence, control torque magnitude, and reaction wheel momentum evolution will be analyzed to assess closed-loop behavior. Particular emphasis will be placed on evaluating disturbance rejection capability and identifying configurations that reduce momentum buildup and the risk of actuator saturation. Comparative analysis across multiple maneuver profiles is expected to provide insight into optimal geometric and control parameter selection.

The anticipated outcome of this work is a validated simulation framework demonstrating that a quaternion-based PD controller, when combined with an appropriately designed reaction wheel geometry, can achieve accurate and robust attitude control under realistic disturbance conditions. Although the proposed controller is based on classical control principles, it is expected to offer performance comparable to more complex methods while retaining simplicity and transparency. The results of this study are intended to serve as a baseline for future extensions, including adaptive control, fault-tolerant strategies, momentum management using magnetic torquers, and hardware- in-the-loop validation.

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